

UNIVERSITY OF MAKATI

College of Computing and Information Sciences OPERSYS

Week 16 Laboratory Activity

Scope: Memory Management (Segmentation) - 35 points

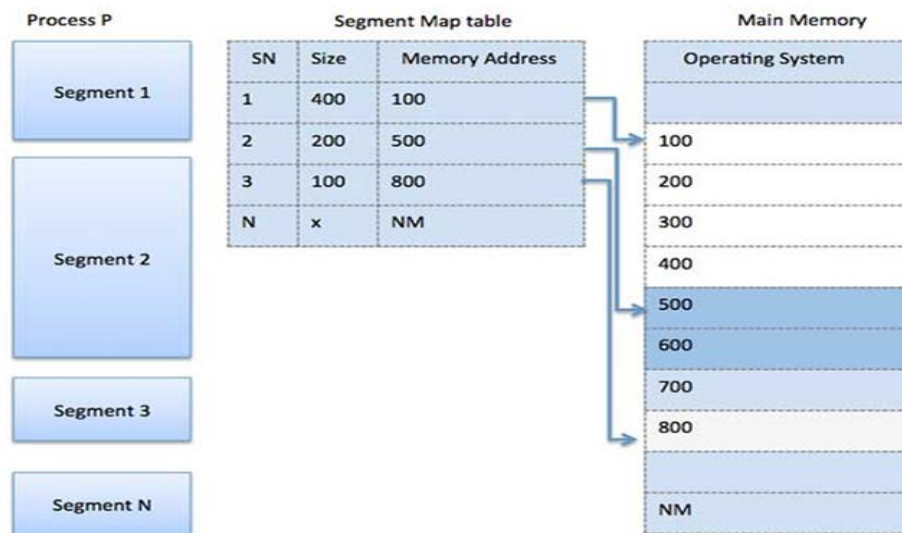


Fig. 1 Segmentation

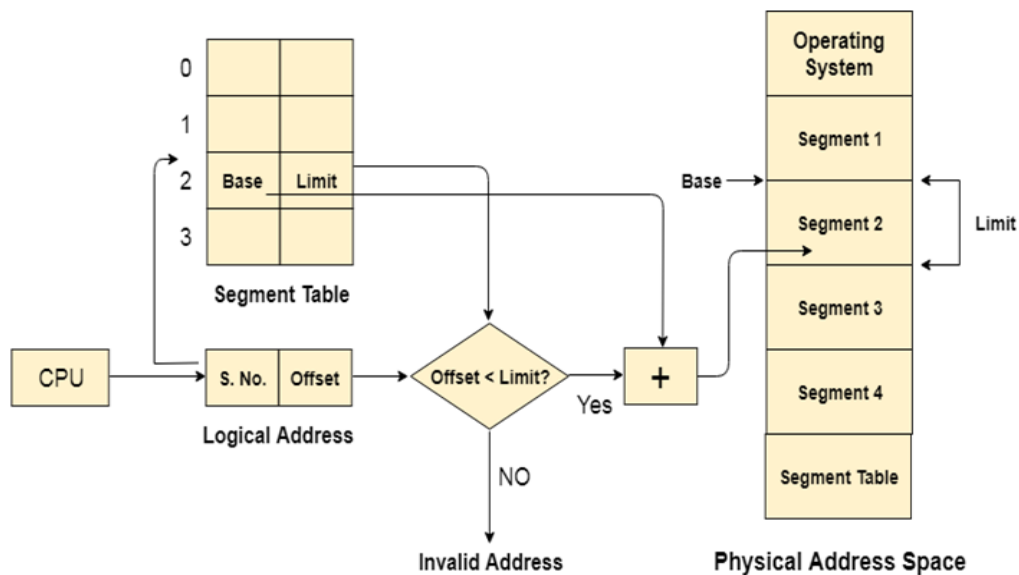


Fig. 2 Translation of Logical address into physical address by segment table

Required:

- 1.) Using the figure below, allocate the processes from the given segments to the main memory.
- 2.) Can you allocate an incoming Process assigned to Segment 5 with a size of 200? Explain.
- 3.) Identify all possible external fragments, after allocating all the segments.

Process P	Segment Map Table			Main Memory
Segment 1	SN	Size	Memory Address	OS
	1	200	100	
	2	500	600	100
Segment 2	3	300	1200	200
	4	100	400	300
	5			400
				500
Segment 3				600
				700
				800
				900
Segment 4				1000
				1100
Segment 5				1200
				1300
				1400
				1500

Fig. 3 Memory Allocation using Segmentation

Rubrics for Problem Solving (Question No.1) 15 points

Problem Solving				
	Excellent 5 pts	Good 4 pts	Average 3 pts	Fair 2 pts
Content	Excellent Appropriate content is used for each problem. Student clearly understands the mathematical concepts.	Good Appropriate content is used for each problem. Student shows some understanding of the mathematical concepts.	Average Appropriate content may be used. Student shows little understanding of the mathematical concepts.	Fair Appropriate content is not observed. Student does not demonstrate an understanding of the mathematical concepts.
Solution/Organization	Excellent The solution is written in clear and coherent way. Solution is presented in a very organized manner.	Good The solution is written in clear and coherent way.	Average The solution is not written in clear and coherent way.	Fair The solution is not written in clear and coherent way, or may not be observed.
Accuracy	Excellent Solution is very clear and accurate.	Good Solution is clear and accurate.	Average Solution is somehow clear and correct.	Fair Solution is not clear and may not be correct.

Rubrics for Essay (Question Nos. 2 & 3) 10 points each

Rubrics for Essay			
	Excellent 5 pts.	Good 3 pts.	Approaching standards 2 pts.
Ideas and Content	What you are writing about is clear and well-expressed, including specific examples to demonstrate what you learned. Well done!	What you are writing about is clear. You answered the question. Some support may be lacking, or your sentences may be a bit awkward. Overall, a decent job.	You put thoughts into this, but there is no real evidence of learning. More specific information is needed or you need to follow the directions more closely.
Sentence Fluency	Sentences are complete and they connect to one another easily when they are read out loud. Your writing 'flows.'	Sentences are complete and able to be understood.	Some sentences are complete and easy to understand. Others require some work.